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Comprehensive Question Bank: Unit 6 — Introduction to Artificial Intelligence (AI) and Machine Learning (ML)

Section A: Multiple-Choice Questions (MCQs)

Topic 6.1.1: What is AI?

1. What does Artificial Intelligence (AI) mean?
   A) A machine that only performs arithmetic
   B) The ability of a machine to perform tasks that normally require human thinking
   C) A computer that stores large amounts of data
   D) A device that only follows fixed electrical signals

2. Who proposed the famous test to measure whether a machine can "think"?
   A) Arthur Samuel
   B) Alan Turing
   C) Francis Galton
   D) Bill Gates

3. What is the name of the test Alan Turing designed in 1950?
   A) The Machine Test
   B) The Turing Test
   C) The Samuel Test
   D) The Intelligence Test

4. In the Turing Test, what must a human judge fail to do for the machine to "pass"?
   A) Type a message
   B) Reliably tell the machine apart from a real human
   C) Understand the machine's code
   D) Switch off the machine

5. Which of the following is a real-life Pakistani example of AI mentioned in the chapter?
   A) A calculator app
   B) A chatbot used by HBL for customer support
   C) A basic alarm clock
   D) A printed newspaper

6. The traffic control system introduced by the Safe City Authority is located in which city?
   A) Karachi
   B) Lahore
   C) Islamabad
   D) Peshawar
7. Urdu speech-to-text apps used in schools are an example of which technology?
   A) Artificial Intelligence
   B) A word processor
   C) A spreadsheet tool
   D) A basic calculator
8. A fixed microwave popcorn button that always cooks for exactly 3 minutes, regardless of the food, is best described as:
   A) Machine Learning
   B) Rule-based AI
   C) Unsupervised Learning
   D) A confusion matrix
9. Which statement about AI is TRUE according to the chapter?
   A) AI does not necessarily mean the machine is conscious
   B) AI always means the machine is thinking exactly like a human
   C) AI can only be used in banking
   D) AI cannot understand speech

Topic 6.1.2: What is ML?

1. Machine Learning (ML) is best described as:
   A) A completely separate field unrelated to AI
   B) A part of AI where machines learn from data
   C) A method of programming every single rule by hand
   D) A device used only for gaming
2. What does ML allow a computer to do?
   A) Follow only pre-written instructions forever
   B) Learn from past data to predict or decide something
   C) Avoid using any data at all
   D) Replace human teachers completely
3. Which of these is an example of ML mentioned in the chapter?
   A) A digital clock showing time
   B) YouTube recommending videos based on watch history
   C) A basic light switch
   D) A printed textbook

4. A weather app predicting tomorrow's temperature using past data is an example of:
   A) Rule-based programming
   B) Machine Learning
   C) A confusion matrix
   D) A fixed algorithm with no data

5. Who coined the term "Machine Learning" while writing a checkers-playing program in 1959?
   A) Alan Turing
   B) Arthur Samuel
   C) Francis Galton
   D) Elon Musk

6. Arthur Samuel's checkers program became better over time because:
   A) It was reprogrammed manually every day
   B) It improved by playing against itself and learning from the results
   C) It had no data at all
   D) It only followed a single fixed rule

Topic 6.1.3: Differences between AI and ML

1. Which statement best summarizes the relationship between AI and ML?
   A) ML and AI are completely unrelated fields
   B) ML is a part of AI; AI is the broader concept
   C) AI is a part of ML
   D) AI and ML are exactly the same thing

2. According to Table 6.1, does AI always need data?
   A) Yes, always
   B) Sometimes — it uses data to act smartly
   C) Never
   D) Only on weekends

3. According to Table 6.1, does ML always need data?
   A) Yes, always — it needs data to learn
   B) No, never
   C) Only sometimes
   D) Only for chatbots

4. Which of the following behaves more like a human (e.g., speaking, seeing, understanding)?
   A) AI
   B) ML
   C) Neither
   D) Both equally in all cases

5. A smart oven that weighs food, checks moisture, and learns how to cook it perfectly over time represents:
   A) Rule-based AI only
   B) Machine Learning
   C) A confusion matrix
   D) A fixed timer system
6. Which of these is the best analogy for the relationship between AI and ML?
   A) AI is the experience; ML is the brain
   B) AI is the brain; ML is the experience that teaches the brain how to improve
   C) AI and ML are two unrelated brains
   D) ML is bigger than AI
7. A chess engine that follows only pre-programmed exact move sequences without learning from games is an example of:
   A) Machine Learning
   B) Rule-based AI
   C) Unsupervised Learning
   D) Supervised Learning

Topic 6.2.1: Supervised Learning

1. Supervised learning is best defined as:
   A) Training a machine with no data at all
   B) Training a machine using labeled data, where both input and correct output are provided
   C) Training a machine using only unlabeled data
   D) A method that requires no human involvement ever
2. In the fruit example, what does labeling a photo as "This is an Apple" provide to the machine?
   A) Nothing useful
   B) A labeled example it can learn from
   C) A random guess
   D) An unsupervised cluster
3. Which of the following best matches the concept of supervised learning?
   A) Sorting unlabeled coins by shape
   B) Studying math with a textbook that has an answer key
   C) Grouping songs with no information at all
   D) Randomly guessing outcomes with no feedback

4. In the 1990s, early spam filters became more effective mainly because:
   A) Spammers stopped sending emails
   B) Millions of users manually labeled emails as "Spam" or "Not Spam"
   C) No data was used at all
   D) Emails were deleted automatically without review

5. A system trained on students' past marks and attendance (each labeled "passed" or "failed") to predict future failure is an example of:
   A) Unsupervised Learning
   B) Supervised Learning
   C) Rule-based AI only
   D) A confusion matrix

6. In the training loop of a supervised learning system, what comes immediately after "Feature Extraction"?
   A) Error Check
   B) Make a Prediction
   C) Adjustment
   D) Collect the Training Data

7. What is a "feature" in the context of supervised learning?
   A) The correct label of an example
   B) A measurable detail of the data, such as color, shape, or size
   C) The final prediction made by the model
   D) The name of the algorithm

Topic 6.2.2: Unsupervised Learning

1. Unsupervised learning is best defined as:
   A) Training with labeled data only
   B) The machine finds patterns or groups in unlabeled data without being told the correct answer
   C) A method that never uses data
   D) The same as supervised learning

2. Sorting a pile of mixed foreign coins into groups by shape and color, without knowing their names, is an analogy for:
   A) Supervised Learning
   B) Unsupervised Learning
   C) A confusion matrix
   D) Linear Regression

3. A school forming student groups based on similar interests, without being told which grouping is "best," is an example of:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Rule-based AI
   D) Accuracy calculation

4. In unsupervised learning, what does the machine NOT receive?
   A) Data
   B) Labels or correct answers
   C) Features
   D) Patterns

5. A store using purchase data to group customers by shopping habits, without knowing an "official" name for each group, demonstrates:
   A) Unsupervised Learning
   B) Supervised Learning
   C) A confusion matrix
   D) Linear Regression

6. In the unsupervised clustering pipeline for online shoppers, what is the step that comes right after "Extract Features"?
   A) Form Clusters
   B) Measure Similarity
   C) Business Uses the Groups
   D) Collect Unlabeled Data

Topic 6.2.3: Supervised vs Unsupervised Learning

1. According to Table 6.3, which type of learning uses labeled data?
   A) Supervised Learning
   B) Unsupervised Learning
   C) Both use labeled data equally
   D) Neither uses data

2. According to Table 6.3, the main task of unsupervised learning is to:
   A) Predict or classify
   B) Group or cluster similar data
   C) Calculate accuracy
   D) Draw a regression line

3. Predicting student results using marks and study hours is an example of:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Neither
   D) A confusion matrix only
4. Grouping customers based on shopping behavior, without predefined categories, is an example of:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Linear Regression
   D) A confusion matrix
5. In a school setting, identifying which specific students need extra help is best matched with:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Neither type of learning
   D) A rule-based system only
6. If a machine "finds patterns on its own" without being told the correct answers, this is:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Linear Regression only
   D) A confusion matrix
7. Identifying spam emails from an inbox using past labeled examples is an example of:
   A) Supervised Learning
   B) Unsupervised Learning
   C) A confusion matrix
   D) K-Means Clustering only
8. Dividing products into unlabeled groups based on shared features is an example of:
   A) Supervised Learning
   B) Unsupervised Learning
   C) Linear Regression
   D) Accuracy calculation

Topic 6.3.1: Smart Assistants and Recommendations

1. Which technologies allow Google Assistant to understand a spoken Urdu command?
   A) Only a microphone
   B) Speech recognition and natural language processing, both powered by AI
   C) A basic calculator function
   D) A confusion matrix

2. Recommendation systems, like the one used by YouTube, primarily work by:
   A) Randomly selecting videos
   B) Suggesting content based on your past activity
   C) Ignoring user history completely
   D) Only showing the newest videos

3. Which of the following apps is given as an example of using AI-based recommendations?
   A) A basic calculator
   B) Daraz, suggesting clothes based on browsing history
   C) A digital clock
   D) A printed catalogue

4. According to the chapter, what specifically helps recommendation systems improve their suggestions over time?
   A) Rule-based AI only
   B) Machine Learning, by analyzing what users like
   C) Random selection
   D) Manual reprogramming for every user

5. Saying "Kal 8 bajay mujhe jagana" to Google Assistant and having it set an alarm demonstrates:
   A) Unsupervised Learning only
   B) AI-powered speech recognition and natural language processing
   C) A confusion matrix
   D) Linear Regression

Topic 6.3.2: Business Use Cases Using Customer Analysis and Chatbots

1. What is the main purpose of customer analysis for businesses using AI?
   A) To slow down business operations
   B) To find out which products are best to sell and understand customer needs
   C) To remove all customer data
   D) To avoid using any technology

2. Cheetay, a grocery delivery app, uses ML to:
   A) Deliver groceries manually with no predictions
   B) Suggest items customers often buy and predict city-wise demand
   C) Avoid using any customer data
   D) Replace all human delivery riders
3. Chatbots used by Pakistani banks and telecom companies are designed to:
   A) Only send advertisements
   B) Understand typed questions and respond automatically without a human agent
   C) Replace all bank employees permanently
   D) Only work during office hours
4. According to the chapter, how does AI help businesses overall?
   A) It slows down operations and increases costs
   B) It helps businesses work faster, save money, and keep customers happy
   C) It only helps with advertising
   D) It has no measurable business benefit

Topic 6.3.3: Bias, Privacy, and Responsibility

1. What is "bias" in AI, as defined in the chapter?
   A) A system that works perfectly for everyone
   B) When a machine learns unfair patterns from the data it was trained on
   C) A type of confusion matrix
   D) A method of encrypting data
2. If an AI system is trained mostly on English or male voices, it may:
   A) Work perfectly for every language and voice
   B) Not work well with Urdu speakers or female voices
   C) Automatically fix its own bias
   D) Stop functioning completely
3. According to the chapter, how can bias in AI be reduced?
   A) By using less data
   B) By using diverse and balanced training data
   C) By ignoring the issue completely
   D) By removing all human oversight
4. Which of the following is listed as a best practice for protecting user privacy?
   A) Collecting as much data as possible
   B) Asking for user permission before collecting data
   C) Sharing data publicly without consent
   D) Avoiding encryption entirely

5. If an AI-powered facial recognition system wrongly accuses someone, who holds the responsibility according to the chapter?
   A) The AI system itself
   B) The humans who created or used the system
   C) No one is responsible
   D) The camera manufacturer only
6. Which of the following is NOT listed as a best practice for privacy protection?
   A) Ask for user permission
   B) Protect data using encryption
   C) Collect as much personal data as possible, regardless of necessity
   D) Do not collect more data than needed
7. What does the chapter say about even top global research teams and bias/privacy issues?
   A) They never make mistakes
   B) They struggle with these exact issues daily
   C) They are not affected by bias at all
   D) They avoid using data entirely

Topic 6.4: Supervised and Unsupervised Algorithms (General)

1. Which algorithm predicts future values based on a straight-line relationship?
   A) Linear Regression
   B) Decision Trees
   C) K-Means Clustering
   D) Confusion Matrix
2. Which algorithm uses a tree-like model to make decisions based on yes/no questions?
   A) Linear Regression
   B) Decision Trees
   C) K-Means Clustering
   D) Accuracy Calculation
3. Which algorithm groups similar items together, such as students with similar interests?
   A) Linear Regression
   B) Decision Trees
   C) K-Means (Clustering)
   D) Confusion Matrix

4. According to the chapter's analogy, an algorithm is best compared to:
   A) A random number generator
   B) A recipe that tells the computer what data to use and what steps to follow
   C) A confusion matrix
   D) A single fixed rule with no steps
5. Choosing the right algorithm mainly depends on:
   A) The programmer's personal preference only
   B) The type of data and the kind of problem being solved
   C) The color of the computer
   D) The time of day

Topic 6.4.1: Linear Regression

1. In the 1800s, who plotted parent and child heights to discover a straight-line predictive relationship?
   A) Alan Turing
   B) Francis Galton
   C) Arthur Samuel
   D) Isaac Newton
2. In simple linear regression predicting test scores from study hours, what is the "Input (X)"?
   A) Test Score
   B) Study Hours
   C) Accuracy
   D) Confusion Matrix
3. In simple linear regression predicting test scores from study hours, what is the "Output (Y)"?
   A) Study Hours
   B) Test Score
   C) Accuracy
   D) Feature Extraction
4. What does linear regression attempt to find in a dataset?
   A) A random scatter of points
   B) A straight line that best fits the data
   C) A confusion matrix
   D) An unlabeled cluster
5. Simple linear regression uses how many independent input variables?
   A) One
   B) Two
   C) Three
   D) As many as needed

6. Which type of regression should be used when predicting an outcome based on multiple factors, such as study hours, attendance, and sleep?
   A) Simple Linear Regression
   B) Multiple Linear Regression
   C) K-Means Clustering
   D) Confusion Matrix
7. Using the equation Score = (5 × Study Hours) + 30, what is the "5" called?
   A) The baseline
   B) The slope
   C) The accuracy
   D) The label
8. Using the equation Score = (5 × Study Hours) + 30, what does "30" represent?
   A) The slope
   B) The predicted score when study hours equal zero (the baseline)
   C) The total number of students
   D) The accuracy percentage
9. Using the equation Score = (10 × Study Hours) + 20, what is the predicted score for a student who studied 5 hours?
   A) 50
   B) 70
   C) 60
   D) 20
10. Which real-world use case is NOT mentioned as an application of linear regression in the chapter?
   A) Predicting sales
   B) Predicting temperature
   C) Sorting unlabeled coins into groups
   D) Predicting student performance

Topic 6.5.1: Confusion Matrix

1. What is a confusion matrix used for?
   A) To store unlabeled data
   B) To show how many predictions/classifications were correct or incorrect
   C) To calculate the slope of a regression line
   D) To translate Urdu speech to text

2. Which historical example is used in the chapter to introduce the confusion matrix concept?
   A) Francis Galton's height studies
   B) World War II radar systems distinguishing aircraft from birds
   C) Arthur Samuel's checkers program
   D) The Turing Test

3. In a confusion matrix predicting pass/fail, what does "True Positive (TP)" mean?
   A) Model predicted fail, and the student failed
   B) Model predicted pass, and the student actually passed
   C) Model predicted pass, but the student failed
   D) Model predicted fail, but the student passed

4. In a confusion matrix predicting pass/fail, what does "True Negative (TN)" mean?
   A) Model predicted pass, and the student passed
   B) Model predicted fail, and the student actually failed
   C) Model predicted pass, but the student failed
   D) Model predicted fail, but the student passed

5. In a confusion matrix predicting pass/fail, what does "False Positive (FP)" mean?
   A) Model predicted fail, and the student failed
   B) Model predicted pass, but the student actually failed
   C) Model predicted pass, and the student passed
   D) Model predicted fail, and the student passed

6. In a confusion matrix predicting pass/fail, what does "False Negative (FN)" mean?
   A) Model predicted pass, and the student passed
   B) Model predicted fail, and the student failed
   C) Model predicted fail, but the student actually passed
   D) Model predicted pass, but the student failed

7. What is the correct formula for Accuracy?
   A) Accuracy = TP / (TP + FN)
   B) Accuracy = (TP + TN) / (TP + TN + FP + FN)
   C) Accuracy = FP / (FP + FN)
   D) Accuracy = TN / (TP + TN)

8. Given TP = 7, TN = 5, FP = 2, FN = 1 for 15 students, what is the accuracy?
   A) 0.70
   B) 0.80
   C) 0.90
   D) 0.60

9. Given TP = 8, TN = 6, FP = 4, FN = 2 for 20 students, what is the accuracy?
   A) 0.60
   B) 0.70
   C) 0.80
   D) 0.90

10. Why is the confusion matrix considered useful for evaluating an ML model?
    A) It replaces the need for any training data
    B) It helps identify exactly where a model is making mistakes and enables accuracy calculation
    C) It only works for linear regression
    D) It eliminates the need for labeled data

Section B: Short Answer Questions (SQs)

Topic 6.1.1: What is AI?

1. Define Artificial Intelligence (AI) in your own words.
2. What question did Alan Turing ask in 1950, and why was it significant?
3. Briefly explain what the Turing Test measures.
4. State two real-life examples of AI used in Pakistan.
5. Identify whether a fixed-timer microwave button is an example of rule-based AI or Machine Learning, and briefly explain why.

Topic 6.1.2: What is ML?

1. Define Machine Learning (ML) in your own words.
2. Explain briefly how Arthur Samuel's checkers program relates to the origin of Machine Learning.
3. State two real-life examples of Machine Learning mentioned in the chapter.
4. Explain briefly how a weather app uses ML to predict tomorrow's temperature.

Topic 6.1.3: Differences between AI and ML

1. Differentiate between AI and ML using at least two points of comparison.
2. Explain briefly why "ML is a part of AI" rather than the two being separate fields.
3. Identify whether a smart oven that learns cooking times over time is an example of AI, ML, or both, and justify your answer.
4. List two features from Table 6.1 that distinguish AI from ML.

Topic 6.2.1: Supervised Learning

1. Define supervised learning in your own words.
2. Explain briefly how the "textbook with an answer key" analogy relates to supervised learning.
3. Identify whether predicting student failure based on labeled past marks is supervised or unsupervised learning, and justify your answer.
4. Briefly explain how labeled data helped improve spam filters in the late 1990s.
5. Map out, in order, the main steps of the supervised learning training loop described in the chapter.
6. What is a "feature" in the context of a supervised learning model? Give one example.

Topic 6.2.2: Unsupervised Learning

1. Define unsupervised learning in your own words.
2. Explain briefly how the "sorting mixed foreign coins" analogy relates to unsupervised learning.
3. Identify whether grouping customers by shopping habits without predefined categories is supervised or unsupervised learning, and justify your answer.
4. Briefly explain how a school might use unsupervised learning to form student groups.
5. Map out, in order, the main steps of the unsupervised clustering pipeline for online shoppers.

Topic 6.2.3: Supervised vs Unsupervised Learning

1. Differentiate between supervised and unsupervised learning using at least two points of comparison.
2. State one real-world use of supervised learning and one real-world use of unsupervised learning, in a school context.
3. Identify whether "grouping playlist songs by tempo" needs supervised or unsupervised learning, and justify your answer.
4. Identify whether "predicting house prices" needs supervised or unsupervised learning, and justify your answer.
5. Explain briefly how you can tell, in general, whether a given scenario needs supervised or unsupervised learning.

Topic 6.3.1: Smart Assistants and Recommendations

1. Define what a smart assistant is, and give one example.
2. Explain briefly how a recommendation system, like YouTube's, works.
3. State two examples of AI-based recommendation systems mentioned in the chapter.
4. Explain briefly what happens technically when you say a command to Google Assistant in Urdu.

Topic 6.3.2: Business Use Cases Using Customer Analysis and Chatbots

1. Define customer analysis and explain how businesses use it.
2. Explain briefly how Cheetay uses Machine Learning in its operations.
3. State two benefits AI provides to businesses, according to the chapter.
4. Explain briefly how chatbots help Pakistani banks and telecom companies.

Topic 6.3.3: Bias, Privacy, and Responsibility

1. Define "bias" in AI, as described in the chapter.
2. Explain briefly why a voice-recognition system trained mostly on male English speakers might fail for other users.
3. List two best practices for protecting user privacy in AI systems.
4. Identify who is responsible when an AI-powered system makes a wrong decision, and justify your answer.
5. Explain briefly why finding bias in an AI system is considered a valuable skill rather than a failure.

Topic 6.4: Supervised and Unsupervised Algorithms

1. Define an algorithm using the "recipe" analogy from the chapter.
2. List the three algorithms mentioned in the chapter, along with one real-world use case for each.
3. Explain briefly what factors determine which algorithm should be chosen for a given problem.

Topic 6.4.1: Linear Regression

1. Define linear regression in your own words.
2. Explain briefly how Francis Galton's height study relates to the discovery of linear regression.
3. Identify the Input (X) and Output (Y) in the example of predicting test scores from study hours.
4. Differentiate between simple linear regression and multiple linear regression.
5. Given the equation Score = (10 × Study Hours) + 20, predict the score for a student who studied 4 hours, and explain your calculation.
6. Explain briefly what the "baseline" value in a linear regression equation represents.

Topic 6.5.1: Confusion Matrix

1. Define a confusion matrix in your own words.
2. Explain briefly how the World War II radar example relates to the concept of a confusion matrix.
3. Differentiate between a False Positive and a False Negative in a pass/fail prediction context.

4. State the formula for calculating accuracy using TP, TN, FP, and FN.
5. Given TP = 6, TN = 4, FP = 2, FN = 3, calculate the total number of predictions and the accuracy.
6. Explain briefly why a model with 80% accuracy might still make dangerous mistakes in some real-world situations.

Section C: Long / Extensive Questions (LQs)

Topic 6.1: AI, ML and Overview

1. Discuss in detail the concept of Artificial Intelligence, including its formal definition, the significance of the Turing Test, and at least three real-life examples of AI used in Pakistan.
   a) Define Artificial Intelligence and explain the historical context of the Turing Test.
   b) Describe three real-life Pakistani examples of AI mentioned in the chapter.
   c) Evaluate whether passing the Turing Test proves that a machine can genuinely "think."

2. Analyze the relationship between Artificial Intelligence and Machine Learning in detail.
   a) Explain, with reference to Table 6.1, how AI and ML differ in definition, goal, data dependency, and human-like behavior.
   b) Using the microwave vs. smart oven analogy, justify why one qualifies as rule-based AI and the other as Machine Learning.
   c) Construct two of your own real-life examples (not from the chapter) — one representing AI broadly and one representing ML specifically — and justify your classification of each.

Topic 6.2: Machine Learning Types

1. Elaborate on Supervised Learning and Unsupervised Learning as the two main categories of Machine Learning.
   a) Define both types of learning and explain the key difference in how they use data.
   b) Construct a step-by-step trace of the supervised learning training loop, from data collection to prediction adjustment, using the fruit classification example.
   c) Construct a step-by-step trace of the unsupervised clustering pipeline using the online shopper grouping example.

2. Design a comparative analysis between Supervised and Unsupervised Learning.
   a) Construct a detailed comparison table covering labeled data requirements, main task, real-world examples, and school-based use cases.
   b) For each of the following scenarios, justify whether supervised or unsupervised learning is the correct approach: (i) detecting fake currency notes, (ii) grouping playlist songs by tempo,

(iii) predicting house prices, (iv) identifying spam emails.
c) Discuss a real-world business or educational scenario where both supervised and unsupervised learning could be used together, and explain how each would contribute.

Topic 6.3: Practical Applications of AI/ML

1. Discuss in detail how AI and ML are applied in daily life through smart assistants and recommendation systems.
   a) Explain how speech recognition and natural language processing work together in a smart assistant, using the Google Assistant Urdu example.
   b) Analyze how recommendation systems like YouTube, Daraz, and Instagram use Machine Learning to personalize user experience over time.
   c) Evaluate the potential downsides or risks of relying heavily on recommendation systems in daily life.
2. Analyze the business applications of AI/ML through customer analysis and chatbots.
   a) Explain how customer analysis, using the Cheetay example, helps businesses make data-driven decisions.
   b) Discuss how chatbots used by Pakistani banks and telecom companies improve customer service efficiency.
   c) Evaluate the trade-offs businesses face when replacing human customer service agents with AI-powered chatbots.
3. Construct a detailed ethical analysis of bias, privacy, and responsibility in AI systems.
   a) Explain, with a real-world style example, how bias can unintentionally enter an AI system through its training data.
   b) Discuss at least three best practices for protecting user privacy in AI-powered applications.
   c) Evaluate who should be held responsible when an AI system causes harm — the developers, the deploying organization, or the AI itself — and justify your position with reasoning from the chapter.
   d) Design a step-by-step bias-auditing procedure that could be used to check any AI system for potential unfairness before it is deployed.

Topic 6.4: Supervised and Unsupervised Algorithms

1. Discuss in detail the three algorithms introduced in the chapter — Linear Regression, Decision Trees, and K-Means Clustering.
   a) Explain the core working principle of each algorithm in your own words.
   b) For each algorithm, describe one real-world business or educational problem it could solve.
   c) Justify which algorithm would be most appropriate for predicting whether a customer will buy a product, and explain your reasoning.

2. Construct a comprehensive explanation and worked demonstration of Linear Regression.
   a) Explain the historical origin of linear regression through Francis Galton's height study.
   b) Using Table 6.5 (Study Hours vs. Test Scores), describe how a best-fit line would be drawn through the data and what it represents.
   c) Given the equation Score = (5 × Study Hours) + 30, calculate the predicted score for students who studied 4, 7, and 9 hours, showing all working.
   d) Differentiate between simple linear regression and multiple linear regression, and explain why multiple linear regression may produce more accurate real-world predictions.

Topic 6.5: Evaluating Model Performance

1. Discuss in detail how a Confusion Matrix is used to evaluate the performance of a Machine Learning model.
   a) Define True Positive, True Negative, False Positive, and False Negative, using the student pass/fail scenario.
   b) Construct a full confusion matrix and calculate the accuracy for a class of 20 students, where TP = 8, TN = 6, FP = 4, and FN = 2. Show all your working.
   c) Analyze which type of error (False Positive or False Negative) would be more serious in a medical diagnostic scenario (e.g., testing for a disease), and justify your reasoning.
   d) Evaluate the limitations of using accuracy alone as a performance metric, and discuss why looking at all four values (TP, TN, FP, FN) individually gives a more complete picture of a model's real-world reliability.